Edinburgh Napier UNIVERSITY | School of Computing, Engineering & the Built Environment

SET08125/SET08825 Artificial Intelligence

Learning Outcomes Covered:	LO2, LO3
Assessment Type:	Report
Overall module assessment	30% coursework, 10% end of unit tests, 60% Exam
For this assessment:	30%
Assessment Limits:	See specification
Submission Deadline:	Friday, 10 April 2026 at 15:00 UK time
Submission Method:	Via Moodle
Turnitin on submissions:	Turnitin report not visible to students
Module Leader:	Ben Paechter
Tutor with direct responsibility:	Dimitra Gkatzia
Return of work and feedback:	Feedback on submissions will normally be provided within three working weeks from the submission date.
Notes:	• You are advised to keep a copy of your submitted assessment. • Please read and follow the 'Fit-to-Sit' guidance if you need to request an extension

Assessment regulations and academic integrity

The University rules on Academic Integrity apply to all submissions. The student academic integrity regulations contain a detailed definition of academic integrity breaches.

- You cannot knowingly permit another student to copy all or part of your work.
- You must not share your work with other students. This includes posting any of your work in any repository that is accessible to others (such as GitHub) and applies also after you have completed the course.
- Asking coursework-related questions in external online forums (such as Stackoverflow) is NOT permitted.

By submitting the report, you are confirming that:

- It is your own work except where explicit reference is made to the contribution of others.
- It has not been submitted for any module, programme or degree at Edinburgh Napier University or any other institution.
- You have not made use of generative Artificial Intelligence (AI) tools.

(a) **Academic skills support:** In advance of submission, you can access the support of the academic skills team. They can help you with any aspect of the assessment that you might struggle with, that is not content related. For example, they can help with time-management, effective reading and note- making, and any aspect of academic writing that you might struggle with. This support is provided through workshops and individual appointments which are bookable online via MyNapier: Improve your Academic & Study Skills (napier.ac.uk). You can also directly email the Academic Skills Adviser, Hannah Awcock, h.awcock@napier.ac.uk for any specific academic skills support you require.

(b) **Use of generative AI:** Please include the Assessment Declaration Cover Sheet in your submission. This is provided below and on moodle. Submissions must be accompanied by a declaration cover sheet to fulfil the requirements of the university Assessment Policy. Note that for this coursework the use of AI is strictly restricted

ASSESSMENT DECLARATION COVER SHEET

Please complete this cover sheet for each assessment submission. For group assessments, each group member must individually complete and submit a cover sheet.

STUDENT REGISTRATION NUMBER: Click or tap here to enter text.

MODULE TITLE: Click or tap here to enter text.

DATE OF SUBMISSION: Click or tap to enter a date.

I declare that, this assessment is my own work and has not been submitted for any other module or degree programme at Edinburgh Napier University or any other institution. This declaration is made in compliance with Edinburgh Napier University's Academic Integrity Regulations.

I have not use Gen AI tools for this submission.

Note: It is also important to check the associated Assessment Brief to understand what is permissible. Unless stated in the Assessment Brief, use of Grammarly for checking spelling and grammar is allowable in this assessment, but it cannot be used to generate content.

Please declare here your use of such tools in this assessment submission:
(Select one of the two options below)

☐ Yes, I have used Gen AI tools for this submission, and I have described how below.

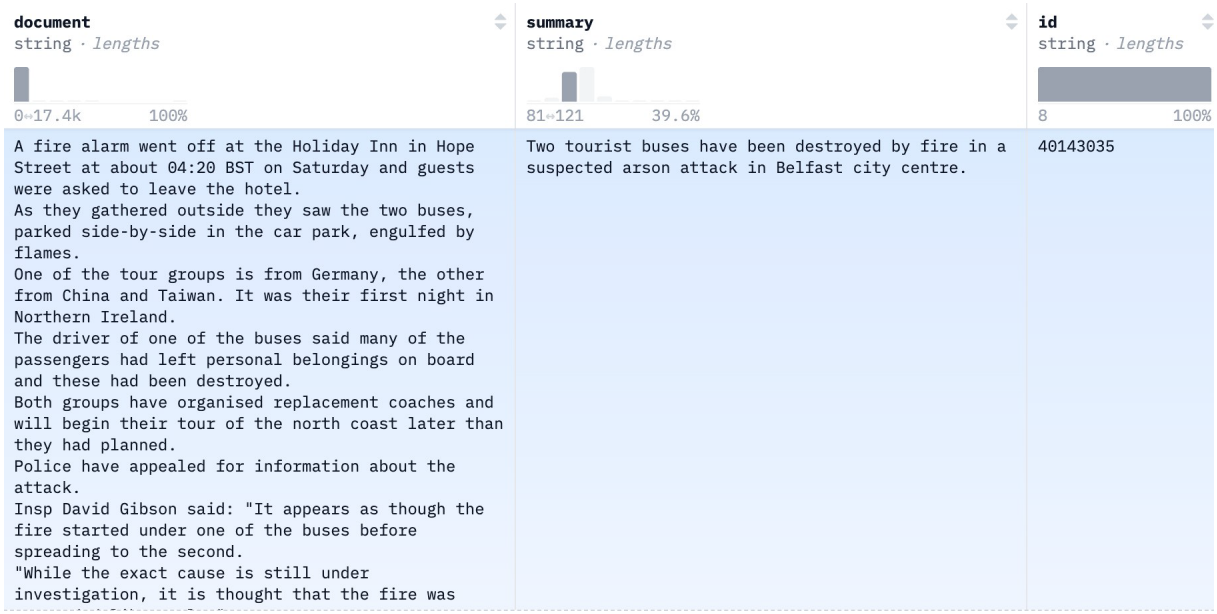
☐ No, I have not used Gen AI tools for this submission.

If you selected **Yes**, briefly describe (in less than 100 words) how you used these tools:

Use	Permitted?	Advice	How to acknowledge use
As a search engine	No	Never ask an AI tool to search for content.	
As an ideas generator/ conversational partner/ debating partner	No	Never use AI as an idea generator.	
To suggest a submission structure	With caution	Consult the assessment brief first to ensure your structure follows the recommendations and meets the learning outcomes.	On the cover sheet
To make suggestions to improve your communication of your ideas	No	Never use AI for suggestions to improve your writing/communication.	
To generate content	No	Never ask an AI tool to generate parts of your submission from scratch. Do not input assessment brief or rubric into AI tools and ask it to generate your submission.	

Assessment Brief - Deadline: 3pm UK time, 10th April on Moodle

You must use the template provided on Moodle. For Part A you will need to compare different approaches to adapting an LLM so that it can solve a summarisation task, as demonstrated by the following dataset: https://huggingface.co/datasets/EdinburghNLP/xsum



document string · *lengths* 0↔17.4k 100%	summary string · *lengths* 81↔121 39.6%	id string · *lengths* 8 100%
A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel. As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland. The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed. Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned. Police have appealed for information about the attack. Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second. "While the exact cause is still under investigation, it is thought that the fire was	Two tourist buses have been destroyed by fire in a suspected arson attack in Belfast city centre.	40143035

The dataset includes articles and corresponding summaries as shown in the image above. You should pick a Large Language Model of your choice (either one shown during tutorials or another suitable model from Hugging Face, considering computational constraints). The dataset follows a standard Hugging Face dataset template. Before applying the learning approaches detailed below in steps 1-3 you will need to: decide how to split the data; define the data processing function; configure the trainer etc.). To help with this, you can refer to Unit 4 tutorial where you fine-tuned models for a similar task.

For Part A, you will need to hand in the Jupyter notebook file evidencing step 1-5 be- low, with all of the code outputs included (**you must follow the template provided on Moodle**). Make sure it can be run via Google Colab. Remember to import and list all required libraries. The contents of step 5 should be included within the same notebook file using dedicated markdown cells. Good Luck!

1. Use zero-shot learning to evaluate the performance of the LLM on this task. Make sure you only test the model on the test set. (5 marks)

2. Use few-shot learning to evaluate the performance of the LLM on this task. Again, test it only on the test set. (5 marks)

3. Finally, employ prompt engineering to solve this task. Again, evaluate only on test set. Focus on carefully articulating instructions, constraints (e.g., length, style), or any special conditions for the summary. Make sure to assess both zero- (3 marks) and few-shot settings (3 marks).

4. For the evaluation, you need to pick an appropriate evaluation metric. Create a table that shows the four different approaches and the results they obtained during the evaluation. (5 marks)

5. Assess and interpret the model performance. Provide a brief discussion concerning Results Interpretation (max 200 words) covering:
   - Findings: Did any method perform significantly better?
   - Surprising errors: Any unexpected outputs or failure cases?
   - Benchmark vs actual performance: Did the results align with expectations?

   (9 marks)

What to hand in on Moodle: A Jupyter notebook file following the **template** provided containing everything. A cover sheet provided on Moodle declaring that this is your own work.